

MAKHACHKALA, Russian Federation

WMO: 374720

Lat: 43.000N

Lon: 47.500E

Elev: -61

StdP: 14.73

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.3	(c) 16.6	(d) 4.8	(e) 7.0	(f) 10.8	(g) 11.4	(h) 9.8	(i) 18.0	(j) 30.2	(k) 35.7	(l) 25.6	(m) 33.8	(n) 5.3	(o) 280	(p) 0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.7	(c) 89.1	(d) 73.4	(e) 86.8	(f) 73.8	(g) 84.8	(h) 73.2	(i) 77.9	(j) 84.9	(k) 76.6	(l) 83.7	(m) 75.1	(n) 82.3	(o) 7.9	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.8	134.4	83.1	74.2	127.2	82.1	72.6	120.5	80.9	41.3	85.1	40.1	84.0	38.6	82.4	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.4	(b) 20.3	(c) 17.6	(e) 6.6	(f) 95.7	(g) 8.5	(h) 3.5	(i) 0.5	(j) 98.3	(k) -4.5	(l) 100.3	(m) -9.2	(n) 102.3	(o) -15.4	(p) 104.9		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			6.3	80.4	8.9	1.8	-0.1	81.7	-5.3	82.7	-10.2	83.8	-16.7	85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	33.7	34.6	41.3	50.4	62.0	71.5	76.6	76.6	68.4	57.3	45.6	37.8
	DBStd	16.72	7.84	7.65	5.10	5.90	5.81	5.04	4.27	4.47	5.39	6.61	7.16	7.55
	HDD50	1834	507	432	272	63	1	0	0	0	0	19	162	378
	HDD65	4831	972	852	734	439	127	8	0	0	27	247	583	842
	CDD50	3570	0	0	3	75	374	644	823	824	553	244	29	1
	CDD65	1095	0	0	0	1	35	203	359	359	130	8	0	0
	CDH74	8191	0	0	0	5	126	1220	3108	3173	549	10	0	0
	CDH80	2148	0	0	0	1	14	252	868	943	70	0	0	0
Wind	WSAvg	7.1	7.0	7.6	8.0	7.7	7.0	6.5	6.2	6.3	7.3	7.2	7.4	7.5
Precipitation	PrecAvg	13.2	0.9	1.1	0.9	0.6	1.0	1.0	1.1	0.9	1.7	1.5	1.5	1.2
	PrecMax	26.1	2.8	2.6	2.5	2.3	3.1	3.0	4.7	3.1	4.8	9.6	4.6	2.9
	PrecMin	8.3	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.0
	PrecStd	3.7	0.7	0.7	0.6	0.5	0.7	0.7	1.0	0.7	1.2	1.8	1.2	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.1	50.7	60.9	72.8	82.3	89.4	92.9	93.6	85.5	75.0	63.7	54.5
		MCWB	43.5	44.1	50.0	58.7	65.9	71.3	72.3	72.2	72.7	64.8	57.5	49.7
	2%	DB	45.7	47.0	54.3	66.7	77.5	85.7	89.0	89.5	81.8	71.5	60.0	51.2
		MCWB	41.8	42.3	47.5	56.6	65.0	70.9	73.2	74.1	71.1	64.6	55.3	47.7
	5%	DB	43.8	44.7	50.8	62.7	74.0	82.9	86.7	87.2	79.1	69.2	57.6	49.3
		MCWB	41.1	41.3	45.5	55.3	64.1	70.1	73.6	74.7	70.1	63.4	54.1	46.5
	10%	DB	42.3	42.9	48.2	59.5	71.3	80.1	84.6	85.0	77.0	67.1	55.4	47.3
		MCWB	40.0	40.3	44.0	53.7	62.9	69.3	73.4	74.3	68.9	61.9	52.0	44.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.5	45.5	51.2	60.4	69.2	76.8	79.2	80.0	76.9	68.1	58.8	51.1
		MCDB	47.0	49.4	59.3	68.6	76.8	84.2	86.5	86.7	82.5	71.7	62.0	53.0
	2%	WB	42.8	43.1	48.5	57.9	66.7	74.0	77.5	78.3	73.6	66.0	56.2	48.5
		MCDB	44.8	45.7	53.5	64.8	74.2	81.8	84.8	85.1	79.1	69.9	58.9	50.5
	5%	WB	41.5	41.8	46.3	55.8	65.3	72.1	76.2	77.1	71.6	64.2	54.3	46.8
		MCDB	43.4	44.0	50.1	61.7	72.3	79.7	83.5	84.1	77.8	68.4	57.2	48.9
	10%	WB	40.2	40.6	44.6	54.0	63.7	70.6	74.8	75.7	69.8	62.5	52.4	44.8
		MCDB	42.0	42.6	47.4	58.9	70.1	78.1	82.1	82.8	75.3	66.6	55.2	47.0
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.4	9.7	11.1	12.5	13.5	13.7	13.7	12.6	12.1	10.8	9.6
		MCDBR	11.4	11.7	15.5	17.2	17.7	17.1	17.0	17.2	14.7	14.9	13.9	11.0
		MCWBR	8.8	8.5	9.4	9.1	8.4	8.4	8.7	9.3	8.0	9.5	10.3	8.4
	5% WB	MCDBR	10.0	10.2	14.0	16.0	15.1	14.6	14.2	15.2	13.4	13.1	12.2	8.8
		MCWBR	8.0	7.7	9.0	8.9	8.3	9.0	9.6	9.8	8.8	9.2	9.5	7.3
Clear-Sky Solar Irradiance	taub		0.368	0.405	0.445	0.509	0.480	0.480	0.512	0.513	0.477	0.423	0.383	0.374
	taud		2.232	2.126	2.064	1.945	2.094	2.153	2.081	2.071	2.127	2.225	2.261	2.221
	Ebn at Noon		233	243	249	242	254	254	244	239	237	235	226	217
	Edh at Noon		30	39	46	56	50	47	50	49	43	35	29	28
All-Sky Solar Radiation	RadAvg		449	696	1023	1437	1755	1954	1921	1721	1321	839	542	399
	RadStd		55	96	89	149	153	112	142	130	123	99	57	48

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page